

District susceptibility: Yes / No answers

Every row is one District + Year. Two independent answers:

(1) Environmental high stress Y/N - from CGWB cohort tertiles.

(2) HMIS observed high burden Y/N - top tertile of official composite burden per 100k.

Source data file: district_susceptibility_profile.csv

Total row

Definitions

- Environmental_susceptibility_high_stress_YN = Yes only if Environmental_stress_band is High (top tertile of Environmental_exposure_index vs all district-years in this table).
- HMIS_observed_high_burden_YN = Yes if HMIS composite burden per 100k is in the top tertile (high) for this dataset.
- Top_GW_stress columns = the three groundwater/monitoring metrics most shifted from cohort average ($|z|$), with signed z in brackets.
- This is not a medical diagnosis and not from regression predictions.

HMIS indicators summed into composite burden

- Childhood Diseases - Severe Acute Malnutrition (SAM)
- Childhood Diseases - Diarrhoea
- Childhood Diseases - Pneumonia
- Childhood Diseases - Measles
- Number of Infant Deaths (1 -12 months) due to Diarrhoea
- Number of Infant Deaths (1 -12 months) due to Pneumonia
- Number of Child Deaths (1 -5 years) due to Diarrhoea
- Number of Child Deaths (1 -5 years) due to Pneumonia
- Number of cases of Infant deaths within 24 hrs of birth
- Number of still births

Full district-year Yes / No table

District	Year	Env Y/N	HMIS Y/N	Env band	HMIS tert	Bur/100k	Exp idx	Top GW 1	Top GW 2	Top GW 3
Araria	2019	No	Yes	Medium	high	69.46	0.1887	Fluoride (F) (z=+2.05)	Monitoring intensity (# wells sampled) ...	Sulphate (SO4) (z=-0.69)
Arwal	2019	No	Yes	Medium	high	126.7	-0.0821	Fluoride (F) (z=+1.44)	Chloride (Cl) (z=-1.36)	Bicarbonate (HCO3) (z=+1.03)
Arwal	2020	No	Yes	Low	high	55.08	-0.5959	Bicarbonate (HCO3) (z=-1.38)	Electrical conductivity (EC) (z=-1.11)	Fluoride (F) (z=-1.01)
Arwal	2021	No	No	Medium	mid	42.52	0.0672	Chloride (Cl) (z=+2.00)	Bicarbonate (HCO3) (z=-1.28)	Monitoring intensity (# wells sampled) ...
Aurangabad	2019	Yes	No	High	mid	38.07	0.4907	Bicarbonate (HCO3) (z=+1.52)	Total dissolved solids (TDS) (z=+1.36)	Electrical conductivity (EC) (z=+1.32)
Aurangabad	2021	Yes	No	High	low	5.67	0.6689	Electrical conductivity (EC) (z=+1.80)	Total dissolved solids (TDS) (z=+1.67)	Nitrate (NO3) (z=+0.86)
Banka	2019	No	No	Medium	mid	31.99	0.1524	Fluoride (F) (z=+2.10)	Bicarbonate (HCO3) (z=-0.64)	Total dissolved solids (TDS) (z=+0.57)
Begusarai	2019	Yes	No	High	mid	40.36	0.2987	Monitoring intensity (# wells sampled) ...	Nitrate (NO3) (z=-0.86)	pH (z=+0.78)
Begusarai	2021	No	No	Medium	low	13.13	0.1032	Sulphate (SO4) (z=+2.69)	Total dissolved solids (TDS) (z=-1.86)	Fluoride (F) (z=-0.99)
Bhagalpur	2019	No	No	Medium	mid	47.14	0.1369	Fluoride (F) (z=+1.48)	Sulphate (SO4) (z=+0.82)	Total dissolved solids (TDS) (z=+0.74)
Bhojpur	2019	Yes	Yes	High	high	48.2	0.356	Monitoring intensity (# wells sampled) ...	Total dissolved solids (TDS) (z=+0.80)	Sulphate (SO4) (z=+0.75)
Bhojpur	2020	No	No	Medium	mid	44.93	0.1507	pH (z=-3.03)	Nitrate (NO3) (z=+2.14)	Sulphate (SO4) (z=+1.88)
Bhojpur	2021	Yes	Yes	High	high	61.9	0.653	Electrical conductivity (EC) (z=+1.97)	Total dissolved solids (TDS) (z=-1.82)	Monitoring intensity (# wells sampled) ...
Buxar	2019	Yes	No	High	low	7.09	0.4456	Fluoride (F) (z=+2.05)	Bicarbonate (HCO3) (z=+1.34)	Monitoring intensity (# wells sampled) ...
Buxar	2020	No	No	Low	low	0.41	-0.6264	pH (z=-2.87)	Bicarbonate (HCO3) (z=-1.94)	Sulphate (SO4) (z=+1.41)
Buxar	2021	No	No	Medium	low	0.94	0.0	Total dissolved solids (TDS) (z=-1.81)	Monitoring intensity (# wells sampled) ...	Bicarbonate (HCO3) (z=+1.23)

Full district-year Yes / No table (continued)

District	Year	Env Y/N	HMIS Y/N	Env band	HMIS tert	Bur/100k	Exp idx	Top GW 1	Top GW 2	Top GW 3
Darbhanga	2019	No	No	Low	low	8.23	-0.6094	Sulphate (SO4) (z=-1.50)	Chloride (Cl) (z=-1.42)	Electrical conductivity (EC) (z=-1.25)
Darbhanga	2021	No	No	Low	mid	25.07	-0.7675	Total dissolved solids (TDS) (z=-1.82)	Sulphate (SO4) (z=-1.64)	Chloride (Cl) (z=-1.25)
Gaya	2019	Yes	Yes	High	high	51.87	0.6096	Bicarbonate (HCO3) (z=+1.98)	Total dissolved solids (TDS) (z=+1.33)	Electrical conductivity (EC) (z=+1.27)
Gaya	2020	Yes	No	High	mid	19.47	0.5616	Nitrate (NO3) (z=+1.95)	Chloride (Cl) (z=+1.29)	Electrical conductivity (EC) (z=+1.13)
Gaya	2021	Yes	No	High	mid	20.77	0.4844	Chloride (Cl) (z=+2.61)	Electrical conductivity (EC) (z=+1.80)	Total dissolved solids (TDS) (z=-1.77)
Gopalganj	2019	No	Yes	Medium	high	78.26	-0.0634	pH (z=-1.05)	Fluoride (F) (z=-0.96)	Sulphate (SO4) (z=+0.91)
Gopalganj	2020	No	No	Low	mid	44.34	-0.5579	Monitoring intensity (# wells sampled) ...	Fluoride (F) (z=-1.01)	Chloride (Cl) (z=-0.88)
Jamui	2019	Yes	No	High	mid	40.73	0.2127	Fluoride (F) (z=+1.48)	Bicarbonate (HCO3) (z=-0.90)	Sulphate (SO4) (z=+0.82)
Jehanabad	2020	Yes	Yes	High	high	65.4	0.48	Sulphate (SO4) (z=+1.94)	Nitrate (NO3) (z=+1.44)	Fluoride (F) (z=-1.01)
Jehanabad	2021	Yes	No	High	mid	45.68	0.738	Chloride (Cl) (z=+2.00)	Total dissolved solids (TDS) (z=+1.24)	Fluoride (F) (z=+1.14)
Katihar	2019	No	No	Medium	low	8.17	0.0009	Chloride (Cl) (z=-0.64)	pH (z=+0.63)	Bicarbonate (HCO3) (z=+0.53)
Katihar	2020	Yes	No	High	low	3.65	0.4352	pH (z=-3.65)	Nitrate (NO3) (z=+3.60)	Electrical conductivity (EC) (z=+2.59)
Khagaria	2019	Yes	Yes	High	high	965.69	0.2787	Chloride (Cl) (z=+2.66)	Sulphate (SO4) (z=-1.40)	Nitrate (NO3) (z=-0.93)
Kishanganj	2019	No	Yes	Medium	high	150.5	-0.1255	Fluoride (F) (z=+2.12)	Bicarbonate (HCO3) (z=-1.19)	Chloride (Cl) (z=-0.88)
Madhepura	2019	No	No	Medium	mid	18.48	0.1387	Fluoride (F) (z=+1.40)	Sulphate (SO4) (z=+1.11)	Bicarbonate (HCO3) (z=-0.98)
Madhubani	2019	No	No	Medium	mid	38.95	-0.0091	Monitoring intensity (# wells sampled) ...	Sulphate (SO4) (z=-1.09)	Fluoride (F) (z=+1.02)

Full district-year Yes / No table (continued)

District	Year	Env Y/N	HMIS Y/N	Env band	HMIS tert	Bur/100k	Exp idx	Top GW 1	Top GW 2	Top GW 3
Munger	2019	Yes	Yes	High	high	438.82	0.2448	Total dissolved solids (TDS) (z=+0.95)	Chloride (Cl) (z=+0.78)	Electrical conductivity (EC) (z=+0.68)
Muzaffarpur	2019	No	No	Low	low	13.83	-0.2972	Electrical conductivity (EC) (z=-0.88)	Sulphate (SO4) (z=-0.81)	Nitrate (NO3) (z=-0.54)
Muzaffarpur	2020	No	No	Low	low	0.87	-0.1692	Monitoring intensity (# wells sampled) ...	Fluoride (F) (z=-1.01)	Bicarbonate (HCO3) (z=+0.98)
Muzaffarpur	2021	Yes	No	High	low	0.29	0.3445	Chloride (Cl) (z=+2.36)	Sulphate (SO4) (z=+1.97)	Total dissolved solids (TDS) (z=-1.83)
Nalanda	2019	No	Yes	Medium	high	88.89	0.154	Monitoring intensity (# wells sampled) ...	Chloride (Cl) (z=+0.44)	Bicarbonate (HCO3) (z=-0.42)
Nalanda	2020	Yes	No	High	low	13.97	0.7367	Nitrate (NO3) (z=+1.48)	Monitoring intensity (# wells sampled) ...	Electrical conductivity (EC) (z=+1.01)
Nalanda	2021	No	No	Medium	low	10.7	0.144	Bicarbonate (HCO3) (z=+1.89)	Total dissolved solids (TDS) (z=-1.79)	Monitoring intensity (# wells sampled) ...
Nawada	2019	No	No	Low	mid	22.4	-0.3777	Sulphate (SO4) (z=-1.49)	Bicarbonate (HCO3) (z=-1.48)	Electrical conductivity (EC) (z=-1.02)
Nawada	2020	No	No	Low	low	4.46	-0.5654	Electrical conductivity (EC) (z=-1.38)	Bicarbonate (HCO3) (z=-1.18)	Sulphate (SO4) (z=-1.13)
Nawada	2021	No	No	Low	low	2.88	-0.4774	Total dissolved solids (TDS) (z=-1.78)	Sulphate (SO4) (z=-1.13)	Electrical conductivity (EC) (z=-0.71)
Patna	2019	No	Yes	Low	high	52.58	-0.5885	Bicarbonate (HCO3) (z=-1.89)	Electrical conductivity (EC) (z=-1.69)	Monitoring intensity (# wells sampled) ...
Patna	2020	Yes	No	High	low	3.05	0.3824	Bicarbonate (HCO3) (z=+2.79)	Monitoring intensity (# wells sampled) ...	Electrical conductivity (EC) (z=+1.78)
Patna	2021	No	Yes	Low	high	65.29	-0.2707	Monitoring intensity (# wells sampled) ...	Bicarbonate (HCO3) (z=-1.18)	Electrical conductivity (EC) (z=-1.08)
Purnia	2020	No	No	Low	mid	30.91	-0.7006	pH (z=-4.12)	Nitrate (NO3) (z=+2.61)	Sulphate (SO4) (z=-1.48)
Rohtas	2019	Yes	No	High	low	1.89	0.4999	Bicarbonate (HCO3) (z=+2.10)	Fluoride (F) (z=+1.90)	Total dissolved solids (TDS) (z=+1.18)
Saharsa	2019	No	Yes	Medium	high	485.83	0.1479	Chloride (Cl) (z=+1.16)	Fluoride (F) (z=+1.03)	Nitrate (NO3) (z=-0.91)

Full district-year Yes / No table (continued)

District	Year	Env Y/N	HMIS Y/N	Env band	HMIS tert	Bur/100k	Exp idx	Top GW 1	Top GW 2	Top GW 3
Samastipur	2019	No	Yes	Medium	high	193.07	-0.0668	Nitrate (NO3) (z=-0.65)	Bicarbonate (HCO3) (z=-0.53)	Chloride (Cl) (z=-0.46)
Samastipur	2020	No	Yes	Low	high	90.72	-0.1459	Nitrate (NO3) (z=+1.74)	Bicarbonate (HCO3) (z=-1.18)	Fluoride (F) (z=-0.98)
Saran	2019	No	No	Medium	mid	19.13	-0.0122	Monitoring intensity (# wells sampled) ...	Chloride (Cl) (z=+0.97)	pH (z=-0.93)
Saran	2020	No	No	Low	low	6.45	-0.2753	Chloride (Cl) (z=-1.25)	Fluoride (F) (z=-1.01)	Bicarbonate (HCO3) (z=+0.63)
Saran	2021	Yes	No	High	low	14.22	0.2128	Nitrate (NO3) (z=+0.92)	Bicarbonate (HCO3) (z=-0.83)	Fluoride (F) (z=+0.65)
Sheohar	2019	No	Yes	Medium	high	166.1	-0.0368	Monitoring intensity (# wells sampled) ...	Fluoride (F) (z=-0.79)	Bicarbonate (HCO3) (z=-0.76)
Sitamarhi	2019	No	Yes	Low	high	106.91	-0.7349	Sulphate (SO4) (z=-1.71)	Electrical conductivity (EC) (z=-1.67)	Chloride (Cl) (z=-1.62)
Sitamarhi	2021	No	No	Low	low	8.68	-1.0018	Electrical conductivity (EC) (z=-2.01)	Total dissolved solids (TDS) (z=-1.85)	Bicarbonate (HCO3) (z=-1.59)
Siwan	2019	No	Yes	Medium	high	115.03	-0.1448	Fluoride (F) (z=-1.01)	Sulphate (SO4) (z=-0.90)	Chloride (Cl) (z=+0.80)
Siwan	2020	No	No	Low	mid	33.03	-0.5985	Monitoring intensity (# wells sampled) ...	Fluoride (F) (z=-1.01)	Chloride (Cl) (z=-0.88)
Siwan	2021	Yes	No	High	mid	36.18	0.2744	Monitoring intensity (# wells sampled) ...	Fluoride (F) (z=+1.37)	Nitrate (NO3) (z=+0.86)
Supaul	2019	No	Yes	Low	high	66.4	-0.3814	Monitoring intensity (# wells sampled) ...	Electrical conductivity (EC) (z=-1.14)	Chloride (Cl) (z=-0.97)
Vaishali	2019	No	No	Low	mid	45.75	-0.3246	pH (z=-1.07)	Sulphate (SO4) (z=-1.00)	Fluoride (F) (z=-0.96)
Vaishali	2020	No	No	Low	low	8.04	-0.3085	Chloride (Cl) (z=-1.25)	Fluoride (F) (z=-1.01)	Electrical conductivity (EC) (z=-0.91)
Vaishali	2021	No	No	Medium	mid	14.33	0.1222	Nitrate (NO3) (z=+1.79)	Fluoride (F) (z=-1.01)	Monitoring intensity (# wells sampled) ...